Arcane Chronicles — Physical data model (MPD)

PostgreSQL · 38 tables · 14 enums · 73 relations · grouped by feature

02. D&D reference data

races

id	PK	uuid
name	NN UQ	varchar
image_url		varchar

classes

id	PK	uuid
name	NN UQ	varchar
image_url		varchar
hit_die	NN	varchar

backgrounds

id	PK	uuid
name	NN UQ	varchar
image_url		varchar

skills

id	PK	uuid
name	NN UQ	varchar
ability	NN	ability_score (enum)

race_skills

race_id	PK FK	uuid
skill_id	PK FK	uuid

class_skills

class_id	PK FK	uuid
skill_id	PK FK	uuid

background_skills

background_id	PK FK	uuid
skill_id	PK FK	uuid

race_abilities

race_id	PK FK	uuid
ability_id	PK FK	uuid

class_abilities

class_id	PK FK	uuid
ability_id	PK FK	uuid
required_level	NN	int

background_abilities

background_id	PK FK	uuid
ability_id	PK FK	uuid

abilities

id	PK	uuid
owner_id	FK	uuid
is_public	NN	boolean
name	NN	varchar
description		text
source	NN	ability_source (enum)
action_type	NN	action_type (enum)
is_spell	NN	boolean
spell_level		int
max_uses		int
created_at	NN	timestamp
updated_at	NN	timestamp

ability_effects

ability_id	PK FK	uuid
effect_id	PK FK	uuid

effects

id	PK	uuid
owner_id	FK	uuid
is_public	NN	boolean
name	NN	varchar
dice_count		int
dice_sides		int
bonus		int
damage_type		varchar
repeat_count		int
description		text
created_at	NN	timestamp
updated_at	NN	timestamp

01. Accounts & campaigns

users

id	PK	uuid
display_name	NN	varchar
email	NN UQ	varchar
password_hash	NN	varchar
created_at	NN	timestamp
updated_at	NN	timestamp

campaigns

id	PK	uuid
gm_id	FK NN	uuid
source_campaign_id	FK	uuid
title	NN	varchar
description		text
image_url		varchar
is_public	NN	boolean
created_at	NN	timestamp
updated_at	NN	timestamp

chapters

id	PK	uuid
campaign_id	FK NN	uuid
title	NN	varchar
position	NN	int
narration		text
image_url		varchar
map_url		varchar
status	NN	chapter_status (enum)
current_scene_id	FK	uuid
created_at	NN	timestamp
updated_at	NN	timestamp

gm_notes

id	PK	uuid
chapter_id	FK NN	uuid
content	NN	text
created_at	NN	timestamp
updated_at	NN	timestamp

scenes

id	PK	uuid
chapter_id	FK NN	uuid
title	NN	varchar
position	NN	int
narration		text
image_url		varchar
map_url		varchar
played_at		timestamp
created_at	NN	timestamp
updated_at	NN	timestamp

06. Live game & combat

games

id	PK	uuid
chapter_id	FK NN	uuid
code	NN UQ	varchar
status	NN	game_status (enum)
started_at		timestamp
ended_at		timestamp
created_at	NN	timestamp

participants

id	PK	uuid
game_id	FK NN	uuid
nickname	NN	varchar
token_hash	NN UQ	varchar
character_id	FK	uuid
status	NN	participant_status (enum)
joined_at	NN	timestamp

combats

id	PK	uuid
game_id	FK NN	uuid
scene_id	FK NN	uuid
status	NN	combat_status (enum)
active_combatant_id	FK	uuid
created_at	NN	timestamp

combatants

id	PK	uuid
combat_id	FK NN	uuid
character_id	FK	uuid
scene_creature_id	FK	uuid
initiative		int

03. Characters & items

characters

id	PK	uuid
campaign_id	FK NN	uuid
source_id	FK	uuid
type	NN	character_type (enum)
name	NN	varchar
image_url		varchar
race_id	FK	uuid
class_id	FK	uuid
level	NN	int
xp	NN	int
is_unique	NN	boolean
alignment		alignment (enum)
background_id	FK	uuid
occupation		varchar
strength	NN	int
dexterity	NN	int
constitution	NN	int
intelligence	NN	int
wisdom	NN	int
charisma	NN	int
worn_armor_id	FK	uuid
max_hp	NN	int
current_hp	NN	int
attitude		attitude (enum)
status	NN	creature_status (enum)
gold	NN	int
is_boss	NN	boolean
biography		text
personality		text
personal_quest		text
secrets		text
gm_notes		text
created_at	NN	timestamp
updated_at	NN	timestamp

item_effects

item_id	PK FK	uuid
effect_id	PK FK	uuid

items

id	PK	uuid
owner_id	FK	uuid
is_public	NN	boolean
source_id	FK	uuid
name	NN	varchar
description		text
image_url		varchar
type	NN	item_type (enum)
price	NN	int
armor_value		int
created_at	NN	timestamp
updated_at	NN	timestamp

character_abilities

id	PK	uuid
character_id	FK NN	uuid
ability_id	FK NN	uuid
uses_remaining		int

character_items

id	PK	uuid
character_id	FK NN	uuid
item_id	FK NN	uuid
quantity	NN	int
for_sale	NN	boolean
unit_price		int

04. Live scene (creatures, trade, chests, riddles, traps)

scene_groups

id	PK	uuid
scene_id	FK NN	uuid
name	NN	varchar
position	NN	int

scene_creatures

id	PK	uuid
scene_id	FK NN	uuid
character_id	FK NN	uuid
group_id	FK	uuid
display_name		varchar
current_hp		int
attitude		attitude (enum)
status		creature_status (enum)
is_revealed	NN	boolean
created_at	NN	timestamp
updated_at	NN	timestamp

scene_creature_items

id	PK	uuid
scene_creature_id	FK NN	uuid
item_id	FK NN	uuid
quantity	NN	int
for_sale	NN	boolean
unit_price		int

chests

id	PK	uuid
scene_id	FK NN	uuid
name	NN	varchar
is_locked	NN	boolean
is_revealed	NN	boolean
created_at	NN	timestamp

chest_items

id	PK	uuid
chest_id	FK NN	uuid
item_id	FK NN	uuid
quantity	NN	int

riddles

id	PK	uuid
scene_id	FK NN	uuid
title	NN	varchar
statement		text
solution		text
is_revealed	NN	boolean
is_solved	NN	boolean
created_at	NN	timestamp

traps

id	PK	uuid
scene_id	FK NN	uuid
name	NN	varchar
description		text
is_revealed	NN	boolean
is_triggered	NN	boolean
created_at	NN	timestamp

trap_effects

trap_id	PK FK	uuid
effect_id	PK FK	uuid

05. Quests & objectives

quests

id	PK	uuid
campaign_id	FK NN	uuid
chapter_id	FK	uuid
scene_id	FK	uuid
related_character_id	FK	uuid
source_id	FK	uuid
title	NN	varchar
description		text
type	NN	quest_type (enum)
status	NN	quest_status (enum)
is_revealed	NN	boolean
position	NN	int
reward_xp	NN	int
reward_gold	NN	int
reward_text		varchar
created_at	NN	timestamp
updated_at	NN	timestamp

quest_reward_items

id	PK	uuid
quest_id	FK NN	uuid
item_id	FK NN	uuid
quantity	NN	int

objectives

id	PK	uuid
quest_id	FK NN	uuid
chapter_id	FK	uuid
label	NN	varchar
narration		text
position	NN	int
target_character_id	FK	uuid
target_riddle_id	FK	uuid
target_chest_id	FK	uuid
target_item_id	FK	uuid
target_trap_id	FK	uuid
target_scene_id	FK	uuid
is_completed	NN	boolean
completed_at		timestamp

Legend

PK primary key · FK foreign key · NN not null · UQ unique · (enum) PostgreSQL enum type

* —— 1 each line goes from the foreign key row (*) to the referenced primary key row (1); line color = feature of the table holding the foreign key

Constraint · quests: MAIN = campaign quest, created automatically with the campaign (always present); the GM only adds its objectives chapter by chapter. SIDE = chapter quest. Constraints in the SQL migration: CHECK (type = MAIN) = (chapter_id IS NULL); unique partial index: one MAIN quest per campaign.

Constraint · objectives: MVP: at most ONE target per objective (CHECK constraint), none for a free objective. The target is informative only: the GM always checks the objective by hand. Chapter progress = completed / total of (objectives of its SIDE quests + MAIN quest objectives with this chapter_id).

Constraint · combatants: Unique character -> character_id; generic creature -> scene_creature_id. Exactly one is set (CHECK constraint).